SOCIAL NETWORKS

NPTEL Online Course • Prof. Sudarshan Iyengar, IIT Ropar

Week 10 — Question Paper (Comprehension-Based MCQ / MSQ)

Comprehension 1 — Bestsellers and the Long Tail

Case Study

An analyst, Farah, studies the catalogue of an online retailer. She ranks every product by how much it sells most popular first and plots sales against this popularity rank. The curve starts very high for the top few products and falls steeply, then stretches far out to the right as a low, thin line over thousands of rarely-bought products. The steep front is the handful of bestsellers; the long thin stretch is the catalogue's many niche items.

A shopkeeper's instinct says to stock only the bestsellers and drop the rest. Farah checks this by comparing two areas under the curve: the sales contributed by the few bestsellers against the sales contributed by the enormous number of niche products added together. She finds the niche stretch, precisely because it is so long, can account for a surprisingly large share of total sales — in many catalogues a clear majority. The same lopsided shape appears far beyond shopping; the frequencies of words in a language, for instance, fall off with rank in just this way.

Farah also asks why some products became bestsellers in the first place. She recalls an experiment in which the very same items were offered to several separate groups of people, with each group shown how often items had already been chosen. Different groups crowned different winners and once an item pulled ahead, the visible head-start drew still more buyers to it. This suggests that early, partly accidental popularity can compound into a large lead, so that the precise ranking at the very top owes as much to cumulative advantage as to intrinsic quality.

Questions 1–10

Q1. [MCQ] What does the “long tail” of Farah's curve refer to, and why does it matter?

- **A.** The few bestsellers, which together make up nearly all of the catalogue's total sales.
- **B.** The many niche products, which together can make up a large share of total sales.
- **C.** A small group of mid-ranked products that sell at exactly the catalogue's average rate.
- **D.** The empty rank positions beyond the last product, which contribute no sales at all.

Answer: **B**

Reference: Lecture 127 – Rich Get Richer: The Long Tail (≈06:00–end)

Solution: The long tail is the vast number of niche, rarely-bought products stretching out to the right; because there are so many of them, their combined sales can be a large share — often the majority — of the total. It is not the bestsellers, not a mid-ranked average group, and not empty positions.

Q2. [MCQ] Why does the shopkeeper's instinct to “stock only the bestsellers” lose money?

A. Because the bestsellers actually sell less in total than any single niche product does.
B. Because the bestsellers stop selling the instant a niche product is taken off sale.
C. Because customers only ever buy products ranked exactly in the middle of the list.
D. Because dropping the niche items discards a stretch that can contribute most sales.

Answer: D

Reference: Lecture 127 – Rich Get Richer: The Long Tail (≈06:00–11:36)

Solution: Although each niche product sells little, the long tail collectively can contribute the larger part of total sales, so cutting it away sacrifices a big slice of revenue. The bestsellers still sell well individually, their sales do not depend on the niche items, and buying is not confined to mid-ranked products.

Q3. [Match the following] Match each term in Column A with its description in Column B.

Column A (term)	Column B (description)
1. Head (bestsellers)	a. word (or item) frequency falls off as 1/rank
2. Long tail	b. early popularity compounds into a widening lead
3. Zipf's law	c. a few products, each selling in very high volume
4. Rich-get-richer	d. very many products, each selling in low volume

A. 1-c, 2-a, 3-d, 4-b
B. 1-d, 2-c, 3-a, 4-b
C. 1-c, 2-d, 3-a, 4-b
D. 1-c, 2-d, 3-b, 4-a

Answer: C (1-c, 2-d, 3-a, 4-b)

Reference: Lectures 126–127 – Rich-get-richer and the long tail

Solution: The head is the few high-volume bestsellers (1-c); the long tail is the many low-volume niche products (2-d); Zipf's law is the 1/rank fall-off of frequency (3-a); and rich-get-richer is the compounding of an early lead (4-b).

Q4. [MCQ] Farah notes that word frequencies follow Zipf's law. What does that say?

A. Every word in the language is used with one and the same fixed frequency throughout.
B. A word's frequency grows in direct proportion to its popularity rank number.
C. Only the single most frequent word is ever used, while all the others go unused.
D. A word's frequency is roughly proportional to 1 divided by its rank.

Answer: D

Reference: Lecture 127 – Rich Get Richer: The Long Tail (≈13:22–end)

Solution: Zipf's law says a word's frequency is roughly proportional to 1/rank — the second-ranked word appears about half as often as the first, the third about a third as often, and so on. Frequencies are not all equal, do not grow with rank, and the lower-ranked words are still used.

Q5. [MCQ] Farah runs the snippet below on the ranked sales. What does share measure?

```
sales = sorted(product_sales, reverse=True)   # sales, most popular first
head = sum(sales[:k])      # the top k 'bestseller' products
tail = sum(sales[k:])      # all the remaining 'niche' products
share = tail / (head + tail)
```

A. The fraction of total sales that comes from the niche (long-tail) products.
B. The fraction of total sales that comes from the top k bestseller products.
C. The number of niche products in the catalogue, divided by the bestsellers.
D. The average price of a niche product relative to a bestseller product.

Answer: A

Reference: Lecture 127 – Rich Get Richer: The Long Tail (≈06:00–end)

Solution: head sums the top-k sales and tail sums everything below them, so tail / (head + tail) is the niche products' portion of total sales — the long tail's share. It is not the bestsellers' share (that would be head over the total), not a count of products, and not a price ratio.

Q6. [Statement type] Consider the two statements about the long-tail catalogue and decide their truth.

Statement I: The long tail contains many niche products, each selling relatively little.
Statement II: The combined sales of the long tail are always negligible next to the bestsellers.

A. Both statements are true.
B. Both statements are false.
C. Statement I is true, but Statement II is false.
D. Statement I is false, but Statement II is true.

Answer: C

Reference: Lecture 127 – Rich Get Richer: The Long Tail

Solution: The first statement is correct: the tail is made of many low-selling niche products. The second is wrong: precisely because there are so many of them, their combined sales are often large — frequently a majority of the total — not negligible. So the first holds and the second fails.

Q7. [MCQ] In the experiment where the same items were shown to several separate groups, what was observed, and what does it imply?

A. Every group crowned the identical winner, showing popularity is fixed purely by quality.
B. Different groups crowned different winners, suggesting cumulative advantage shapes the top ranks.
C. No item was ever chosen by any group, showing visible popularity counts deter all buyers.
D. The groups always reversed the true ranking, proving popularity is the opposite of quality.

Answer: B

Reference: Lecture 126 – Rich Get Richer: A Possible Reason (≈04:49–end)

Solution: Different groups produced different winners, and an early lead drew still more choices, so the top of the ranking reflects cumulative advantage (and some luck), not quality alone. The winners were not identical, items were chosen, and the ranking was not systematically reversed.

Q8. [MCQ] How does the long-tail / rich-get-richer pattern relate to power law from the previous chapter?

A. They are unrelated; sales curves never resemble a network's degree distribution at all.
B. The long tail is a perfectly symmetric bell curve, the exact opposite of a power law.
C. The two contradict each other, since a power law forbids any product from selling well.
D. The sales-versus-rank curve is power-law-like, driven by the same compounding force.

Answer: D

Reference: Lectures 119, 127 – Rich-get-richer and the long tail

Solution: The steep-then-trailing sales curve has the same heavy-tailed, power-law-like shape as a degree distribution, and the same rich-get-richer mechanism drives both. It is not unrelated, not a symmetric bell curve, and the two ideas reinforce rather than contradict each other.

Q9. [MSQ] Which statements about the long tail and rich-get-richer are correct? (Select all that apply.)

I. The niche products are many, but each sells only a little.
II. The long tail's combined sales can exceed those of the bestsellers.
III. Showing how popular items already are can bias new buyers toward them.
IV. Stocking only the bestsellers is always the most profitable strategy.

A. I, II, IV
B. II, III, IV
C. I, II, III
D. I, III

Answer: C (statements I, II, III)

Reference: Lectures 126–127 – Rich-get-richer and the long tail

Solution: Many low-selling niche items (I), a long tail that can outsell the bestsellers in aggregate (II), and popularity cues biasing new buyers (III) are all correct. Because the tail can dominate sales, stocking only bestsellers is not always best, so IV is false.

Q10. [MCQ] Farah summarises the long-tail phenomenon in one sentence. Which is it?

A. Popular products are few but each sells a lot, while niche products are many but each sells little.
B. Popular products are many and niche products are few, so the catalogue is dominated by hits.
C. Every product sells the same amount, so the ranking by popularity carries no information.
D. Niche products each outsell every bestseller, which is why bestsellers are eventually dropped.

Answer: A

Reference: Lecture 127 – Rich Get Richer: The Long Tail (≈11:36–end)

Solution: The long tail is exactly this contrast: a few popular products selling in high volume, set against very many niche products each selling little but together commanding a large share. The other options invert the head/tail sizes, deny any ranking, or wrongly claim each niche item beats the bestsellers.

Comprehension 2 — Modelling an Outbreak Across a Network of Farms

Case Study

An epidemiologist models a contagious livestock disease spreading across a network of farms linked by animal-transport routes. Two ingredients drive the model: the pathogen, summarised by a probability p that an infected farm passes the disease along any one transport link, and the network itself, whose density decides how easily the disease travels. As a first approximation she treats the network as a tree in which each farm trades with about k others, and asks how the outbreak grows level by level.

Across one link the transmission is a coin toss: the infection crosses with probability p and fails with probability 1 − p. Since an infected farm has about k outgoing links, the count it infects is a sum of k such tosses, whose expected value is the key quantity the model watches a single number summarising how many fresh infections one infected farm typically produces. The epidemiologist explains that this number sits on a knife-edge: on one side the outbreak fades out for certain, while on the other it can grow into an epidemic, and public-health measures work precisely by nudging it across that edge — lowering either the per-link transmission risk or the number of trading partners a farm has.

Moving from the tree to a realistic network, she uses the SIR model, in which each farm is Susceptible, then Infected for a fixed infectious period, then Recovered or Removed (immune, or culled and gone). A variant, the SIS model, sends a farm from Infected back to Susceptible, so it can be struck again. She also reframes the whole stochastic process statically: rather than stepping through time, decide each link's fate up front by one coin toss, then read off which farms the infection reaches. The day-by-day simulation steps through time instead:

```
import random
infected = {seed};  recovered = set()
for day in range(T_I):                  # one infected farm stays infectious for T_I days
    newly = set()
    for u in infected:
        for v in G.neighbors(u):
            if v not in recovered and random.random() <= p:
                newly.add(v)            # the coin toss crosses this link
    infected |= newly
# after T_I days an infected farm moves on (recovered/removed, or back to susceptible)
```

Questions 1–10

Q1. [MCQ] In the tree approximation, level 1 holds 10 farms and every farm trades with 10 farms at the next level. How many farms sit at level 5?

- **A.** 10^5, ten times more farms at each successive level.
- **B.** 5^10, swapping the base and the exponent over.
- **C.** 10 × 5 = 50, ten farms on each of the five levels.
- **D.** 10^10, squaring the exponent at the last level.

Answer: A

Reference: Lecture 131 – Simple Branching Process (Continued) (≈00:09–03:06)

Solution: Level 1 holds 10 farms and each farm has 10 children, so every level is ten times the one before: level i holds 10^i farms, giving 10^5 at level 5. Swapping base and exponent, a flat 10 × 5, and 10^10 all misapply the level-by-level multiplication.

Q2. [MCQ] In the tree approximation, what is the expected number of farms that one infected farm directly infects?

A. p + k, the infection probability added to the number of transport links.
B. p × k, the per-link probability times the number of links the farm has.
C. k ÷ p, the number of links divided by the infection probability.
D. 1 − p, the probability that the infection fails to cross a single link.

Answer: B

Reference: Lecture 130 – Simple Branching Process (≈13:53–end)

Solution: Each of the k links transmits independently with probability p, so the expected number of direct infections is the sum of k such chances, p + p + … (k times) = p × k. It is not a sum p + k, not k ÷ p, and 1 − p is the per-link failure probability.

Q3. [MCQ] The single number p × k is the basic reproductive number R_0. A disease has p = 0.25 and k = 40. What is R_0, and what does it imply?

A. R_0 = 0.5, below 1, so the outbreak is certain to die out before long.
B. R_0 = 1, exactly on the edge, so the outbreak neither grows nor fades away.
C. R_0 = 40, equal to k, so every farm is infected within just a single day.
D. R_0 = 10, well above 1, so the outbreak can grow into an epidemic.

Answer: D

Reference: Lecture 132 – Basic Reproductive Number (≈03:25–end)

Solution: R_0 = p × k = 0.25 × 40 = 10, which is well above 1, so each infected farm produces many secondary infections and the outbreak can become an epidemic. The values 0.5, 1, and 40 misuse the formula or confuse R_0 with k.

Q4. [Arrange in order] Arrange the steps of one SIR simulation pass in the correct order.

Steps (in scrambled order):

(i) For each infected farm, toss a coin (probability p) along each link to a susceptible neighbour.
(ii) Seed the network by marking one farm as infected.
(iii) Move farms that have finished their infectious period to recovered/removed.
(iv) Add the neighbours whose coin toss succeeded to the infected set.
(v) Repeat until no infected farms remain, then report which farms were hit.

A. ii → iv → i → iii → v
B. i → ii → iv → iii → v
C. ii → i → iv → iii → v
D. ii → i → iii → iv → v

Answer: C (ii → i → iv → iii → v)

Reference: Lecture 134 – SIR and SIS Spreading Models

Solution: Seed one infected farm (ii), toss the coin along each link from infected to susceptible farms (i), add the successful neighbours to the infected set (iv), retire farms past their infectious period to recovered/removed (iii), and repeat until no infected farms are left, then report the result (v).

Q5. [MCQ] The modeller computes the value below. What does epidemic tell her?

```
R0 = p * k          # p: infection prob per link;  k: links per farm
epidemic = R0 > 1
```

A. Whether the outbreak tends to grow ($R_0 > 1$) rather than die out ($R_0 < 1$).
B. The exact number of farms that will end up infected by the end of the outbreak.
C. The number of days the infectious period T_I lasts for each infected farm here.
D. Whether the network is connected enough for any transport link to exist at all.

Answer: A

Reference: Lecture 132 – Basic Reproductive Number (≈00:26–03:25)

Solution: epidemic is true exactly when $R_0 = p \times k$ exceeds 1, the threshold separating an outbreak that tends to grow from one that dies out. It is not a head-count of eventual infections, not the infectious period, and not a connectivity test.

Q6. [Statement type] Consider the two statements about the SIR and SIS models and decide their truth.

Statement I: An SIR outbreak on a finite network always eventually comes to a stop.
Statement II: An SIS outbreak is likewise guaranteed to stop after finitely many steps.

A. Both statements are true.
B. Both statements are false.
C. Statement I is true, but Statement II is false.
D. Statement I is false, but Statement II is true.

Answer: C

Reference: Lecture 135 – Comparison between SIR and SIS

Solution: The first statement is correct: under SIR, infected farms keep moving permanently into the recovered/removed state, so the supply of reinfectable farms runs out and the process must end. The second is wrong: under SIS farms return to susceptible and can be reinfected, so the outbreak can cycle on indefinitely. So the first holds and the second fails.

Q7. [MCQ] What is the key difference between the SIR and SIS models?

A. SIR sends a recovered farm straight back to susceptible; SIS keeps it infected forever.
B. SIR has a recovered/removed state (immune or culled); SIS returns the farm to susceptible.
C. SIR ignores the infectious period entirely, whereas SIS infects every farm on day one.
D. SIR applies only to trees and SIS only to complete graphs; otherwise they are identical.

Answer: B

Reference: Lecture 134 – SIR and SIS Spreading Models (≈09:23–end)

Solution: In SIR a farm ends up Recovered or Removed (immune, or culled and gone for good), while in SIS it goes from Infected back to Susceptible and can catch the disease again. SIR does not loop back to susceptible, both use an infectious period, and neither is restricted to a particular graph shape.

Q8. [MCQ] Public-health measures aim to push R_0 across its threshold of 1. Which actions do this, and how?

A. Raising p through more transport, or raising k by merging farms, to drop R_0 below 1.
B. Cutting p via biosecurity, or cutting k by restricting movement, to push R_0 under 1.
C. Leaving p and k untouched, since R_0 cannot be changed once the outbreak has begun.
D. Cutting p and k only after R_0 has already fallen below 1 on its own accord.

Answer: B

Reference: Lecture 132 – Basic Reproductive Number (≈06:25–end)

Solution: Since $R_0 = p \times k$, lowering either factor lowers R_0: biosecurity reduces the per-link probability p, and movement restrictions (quarantine) reduce the number of contacts k. Pushing R_0 below 1 turns a would-be epidemic into one that dies out. Raising p or k does the opposite, and R_0 can certainly be influenced during an outbreak.

Q9. [MCQ] In the percolation reformulation, how is the spread decided without stepping through time?

A. Pre-toss a coin per link (open with prob p); a farm is hit if an open path reaches it.
B. Toss one coin per farm in advance; a farm is hit only if its own coin comes up heads.
C. Open every link with certainty, so that every farm in the network is always infected.
D. Remove all links first, so that no farm beyond the seed can ever be reached.

Answer: A

Reference: Lecture 137 – Percolation model (≈06:30–end)

Solution: Percolation fixes each link's outcome up front — open with probability p, closed otherwise — and a farm is infected exactly when an all-open path connects it to the seed, removing the need to track time. The coin is tossed per link (not per farm), links are not all open, and not all removed.

Q10. [MSQ] Which statements about the epidemic models are correct? (Select all that apply.)

I. The basic reproductive number is $R_0 = p \times k$.
II. If $R_0 < 1$ the outbreak dies out with certainty.
III. An SIS outbreak can keep running indefinitely.
IV. The percolation view tracks the spread day by day through time.

A. I, II, IV
B. II, III, IV
C. I, III
D. I, II, III

Answer: D (statements I, II, III)

Reference: Lectures 132–137 – R_0, SIR/SIS, and percolation

Solution: $R_0 = p \times k$ (I), an R_0 below 1 means certain die-out (II), and an SIS outbreak can cycle on forever (III). The percolation view deliberately removes the time dimension — deciding links up front — so IV is false.
